

# YAJNASENI DUTTA

## EMPLOYMENT

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- Assistant Professor (Universitair Docent)**, Universiteit Leiden Sep. 2023–  
Mathematical Institute (tenured Nov. 2024)
- Postdoc**, Rheinische Friedrich-Wilhelms-Universität Bonn Sep. 2022–Aug. 2023  
Complex Geometry Group of Prof. Daniel Huybrechts
- Hausdorff Postdoc**, Rheinische Friedrich-Wilhelms-Universität Bonn Sep. 2019–Aug. 2022  
Complex Geometry Group of Prof. Daniel Huybrechts

## SELECTED RESEARCH VISITS

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- Visitor (ten days) Host: Prof. Daniel Huybrechts Oct. 2023  
Universität Bonn
- Visiting Scholar (three months) Host: Prof. Nero Budur Oct. 2019–Dec. 2019  
Katholieke Universiteit Leuven

## EDUCATION

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- Northwestern University, USA** Sep. 2014–Aug. 2019  
Ph.D. Mathematics **Advisor:** Prof. Mihnea Popa
- University of Illinois at Chicago, USA** Aug. 2013–Aug. 2014  
Ph.D. candidate Mathematics **Advisor:** Prof. Mihnea Popa
- Chennai Mathematical Institute, India** Aug. 2011–Jul. 2013  
M.Sc., Mathematics
- Chennai Mathematical Institute, India** Aug. 2008–Jul. 2011  
B.Sc., Mathematics and Computer Science

## RESEARCH INTERESTS

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Hyperkähler manifolds, generic vanishing theory, birational geometry, Hodge theory and Hodge module theory

## LIST OF PUBLICATIONS/PREPRINTS

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9. **Relative compactified Prym and Picard fibrations associated to very good cubic fourfolds.** jt. with L. Marquand, to appear in Proc. Amer. Math. Soc. <https://arxiv.org/abs/2502.21301>
8. **Twists of intermediate Jacobian fibrations.** jt. with D. Mattei, and E. Shinder preprint available at <https://arxiv.org/abs/2411.01953v1>
7. **Some density results for hyperkähler manifolds.** jt. with E. Izadi, L. Kamenova, and L. Marquand to appear in Math. Res. Lett. <https://arxiv.org/abs/2403.04868>
6. **On symplectic birational self-maps of projective hyperkähler manifolds of  $K3^{[n]}$ -type.** jt. with D. Mattei and Y. Prieto-Montañez *International Mathematics Research Notices*, vol. 5 (2024) DOI:10.1093/imrn/rnae112
5. **Maximal variation of curves on  $K3$  surfaces** jt. with D. Huybrechts *Tunisian J. Math.* vol. 4, 2022, no. 3 DOI:10.2140/tunis.2022.4.443
4. **Generic Vanishing, 1-forms, and Topology of Albanese Maps** jt. with F. Hao and Y. Liu. *Math. Zeitschrift*. Volume 306, no. 56, (2024) DOI:10.1007/s00209-024-03438-3
3. **Vanishing for Hodge ideals on toric varieties** *Mathematische Nachrichten*. vol. 293, 2020. DOI:10.1002/mana.201800363
2. **Effective generation and twisted weak positivity of direct images** jt. with T. Murayama. *Algebra and Number theory*. vol. 13, no. 2, 2019. DOI:10.2140/ant.2019.13.425
1. **On the effective freeness of the direct images of pluricanonical bundles** *Annales de l'Institut Fourier*, vol. 70, no. 4, 2020 . DOI:10.5802/aif.3351

## GRANTS, AWARDS, FELLOWSHIPS AND RECOGNITION

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- Starter budget** for hiring a PhD student (€240,000), Universiteit Leiden Sep. 2023

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| <b>Best Thesis in Mathematics</b> , Northwestern University                                                                                                                                                       | May 2019  |
| <b>Gelfand Award</b> , Northwestern University<br>awarded to the Ph.D. student who has contributed most to the department                                                                                         | May 2018  |
| <b>University Fellowship</b> , University of Illinois at Chicago<br>nominated by the Maths Department among the top 15% of newly recruited graduate students, receiving a full year's stipend and tuition waiver. | 2013–2014 |
| <b>Scholarship for Higher Education</b> , Department of Science and Technology, Govt. of India<br>a component of the programme “Innovation in Science Pursuit for Inspired Research (INSPIRE)”                    | 2008–2011 |
| Qualified <b>regional mathematical olympiad</b> , West Bengal<br>unique track for being part of the IMO-team, India and is held across regions throughout India                                                   | 2007      |

## THESIS SUPERVISION

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| PhD Thesis (co-advising with Prof. Evgeny Shinder): <i>Céline Fietz</i> , Universiteit Leiden                                                            | Dec. 2023–          |
| Master's Thesis: <i>Matthijs Buijsen</i> , <b>Moduli of Sheaves on Threefolds</b><br>Universiteit Leiden                                                 | Sep. 2025–          |
| Master's Thesis (co-advising with Dr. E. Brakkee): <i>Jorge Macein</i> , <b>Twisted Elliptic Fibrations of the Fermat quartic</b><br>Universiteit Leiden | Mar. 2025–          |
| Master's Thesis (co-advised with Prof. E. Shinder): <i>Céline Fietz</i> , <b>Categorical Resolutions of A2 Singularities</b><br>Universität Bonn         | Oct. 2022–Oct. 2023 |
| Bachelor's Thesis: <i>Lucio Verrijzer</i> , <b>The Morita Equivalence and (Graded) Brauer Groups</b> .<br>Universiteit Leiden                            | Feb.–Jun. 2024      |

## TEACHING

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| Instructor: <i>Representation Theory</i><br>Universiteit Leiden.                                                                                 | Spring 2025            |
| Instructor: <i>Advanced Algebraic Geometry: Moduli of sheaves</i><br>national mastermath program, Vrije Universiteit Amsterdam.                  | Fall 2024              |
| Instructor: <i>Mathematical Reasoning</i> Leiden University College.                                                                             | Spring 2024            |
| Instructor: <i>Algebraic Curves</i> Universiteit Leiden.                                                                                         | Fall 2023, 25          |
| Teaching Assistant: <i>Algebraic Geometry II</i> Universität Bonn.                                                                               | Summer 2022            |
| Teaching Assistant: <i>Algebraic Geometry I</i> Universität Bonn.                                                                                | Winter 2022            |
| Hauptseminar Algebra: <i>Klassische Algebraische Geometrie</i> Universität Bonn.                                                                 | Winter 2022            |
| Lecture course: <i>Selected Topics in AG: Moduli of Vector Bundles</i> Universität Bonn.                                                         | Winter 2021            |
| Graduate Seminar: <i>Algebraic Curves and Abelian Varieties</i> Universität Bonn.                                                                | Winter 2020            |
| Teaching Assistant: <i>Calculus (all levels), Linear Algebra, Foundations of Higher Math, Intro to Topology, Graph Theory</i> Northwestern Univ. | Fall 2015–Spring 2018  |
| Grader: <i>Differential Topology, Algebraic Geometry</i> Northwestern Univ.                                                                      | Fall 2014, Winter 2015 |

## INVITED WORKSHOP VISITS

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| <i>Complex Analysis</i><br>Mathematisches Forschungsinstitut Oberwolfach       | Apr. 2026 |
| <i>Geometry of HyperKähler Varieties</i><br>IASM Hangzhou                      | Sep. 2023 |
| <i>Women in Mathematical Physics II</i><br>Banff, Canada (participated online) | Aug. 2023 |

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| Arbeitsgemeinschaft : <i>Twistor D-Modules and the Decomposition Theorem</i><br>Mathematisches Forschungsinstitut Oberwolfach<br><i>Oberwolfach report</i> | Apr. 2023        |
| <i>Algebraic Geometry: Moduli Spaces, Birational Geometry and Derived Aspects</i><br>Mathematisches Forschungsinstitut Oberwolfach                         | Jul. 2022 & 2026 |
| Summer School: <i>Irregular Riemann-Hilbert Correspondence</i><br>Centre Paul Langevin in Aussois                                                          | Sep. 2021        |

## SELECTED INVITED TALKS

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### Conference talks

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| <i>Cubic Fourfolds, Gushel-Mukai Fourfolds, and Hyperkahler Manifolds</i> , ICERM, USA               | Oct. 2026 |
| <i>Higher Dimensional Varieties</i> , Erdős Center, Budapest                                         | Oct. 2026 |
| <i>Rationality problems, Cremona groups, and motivic invariants</i> , LMS symposium, Sheffield       | Jul. 2026 |
| <i>Hodge Theory</i> , Ashoka University, Delhi                                                       | Jan. 2026 |
| <i>Conference in Algebraic Geometry via Eurostar</i> , Leiden, Netherlands                           | Sep. 2025 |
| <i>Japanese European Symposium on Symplectic Varieties and Moduli Spaces</i> , Netherlands           | Sep. 2025 |
| <i>HK days in Sheffield</i> , Sheffield                                                              | Feb. 2025 |
| <i>D-modules, local systems and applications</i> , CRM Montreal                                      | Sep. 2024 |
| <i>GQT symposium</i> , Nijmegen                                                                      | Aug. 2024 |
| <i>DIAMANT symposium</i> , Utrecht                                                                   | Apr. 2024 |
| <i>GQT section: Netherlands Mathematical Congress</i> , Lunteren                                     | Apr. 2024 |
| <i>Fano and Hyperkähler varieties</i> , Genova                                                       | Jan. 2024 |
| <i>Belgian–Dutch Algebraic Geometry Meeting</i> , Leiden                                             | Oct. 2023 |
| <i>Geometry of HyperKähler Varieties</i> , Hangzhou                                                  | Sep. 2023 |
| <i>Derived categories, Hodge theory and Singularities</i> , Sheffield                                | Jun. 2023 |
| <i>Algebraic Geometry section: DMV-Jahrestagung</i> , FU Berlin                                      | Sep. 2022 |
| <i>Workshop: Curves and K3 surfaces</i> , Bonn                                                       | May 2022  |
| <i>D-Modules: Applications to Algebraic Geometry, Arithmetic and Mirror Symmetry</i> , CIRM (Luminy) | Apr. 2022 |
| <i>Singularities in the Midwest Workshop</i> , Wisconsin (Online)                                    | May 2021  |
| <i>AMS Fall Eastern Sectional, “Algebraic Singularities in Arbitrary Characteristic”</i> , (Online)  | Oct. 2020 |
| <i>Algebra, Algebraic Geometry and Number Theory section: DMV-Jahrestagung</i> , Chemnitz (Online)   | Sep. 2020 |

### Lecture Series

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|-----------------------------------------------------------------------------|-----------|
| Swiss-French Workshop in Algebraic Geometry Charmey, Switzerland            | Jan. 2026 |
| D-modules in Bochum – lecture series (3 hours), Ruhr-Universität Bochum     | Mar. 2025 |
| D-modules in birational geometry – Short course (3 hours), Univ. of Chicago | Nov. 2018 |

### Seminars

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|----------------------------------------------------------------|-----------|
| Harvard–MIT Algebraic Geometry Seminar                         | Sep. 2024 |
| Séminaire “Arithmétique et géométrie algébrique” de Strasbourg | Mar. 2024 |
| Algebraic Geometry Seminar, Princeton University               | Feb. 2024 |
| Algebraic Geometry Seminar, USTC, Hefei                        | Sep. 2023 |
| Seminar: ESAGA group joint oberseminar, Essen                  | Jul. 2023 |

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| Mathematics Colloquium, Institute of Mathematical Sc., Chennai                      | Aug. 2022                      |
| Algebraic Geometry Seminar, Columbia University                                     | Mar. 2022                      |
| Algebraic Geometry Seminar, Stony Brook University (Online)                         | Sep. 2021                      |
| Algebraic Geometry Seminar, Princeton University (Online)                           | Jul. 2021                      |
| Online Algebraic Geometry Seminar, UC, San Diego, (Online)                          | Apr. 2021                      |
| Edinburgh Hodge seminar (online)                                                    | Jan. 2021                      |
| Algebra and Geometry Seminar, Frankfurt (Online)                                    | Dec. 2020                      |
| Zoom seminar in TIFR, Mumbai (Online)                                               | Nov. 2020                      |
| Oberseminar Algebraic Geometry, Hannover (Online)                                   | Jul. 2020                      |
| Seminar on Applications of Hodge modules to birational geometry, HU Berlin, Germany | Jan. 2020                      |
| Oberseminar “Algebra und Theoretische Mathematik”, Chemnitz, Germany                | Nov. 2019                      |
| Seminar Number Theory and Algebraic Geometry, KU Leuven, Belgium                    | Oct. 2019                      |
| Algebraic Geometry and Commutative Algebra Seminar, University of Notre Dame        | May 2019                       |
| Algebraic Geometry Seminar, Johns Hopkins, Baltimore                                | Mar. 2019 & Sep. 2024          |
| Algebraic Geometry Seminar, Univ. of Illinois, Urbana Champaign                     | Nov. 2018                      |
| Algebraic geometry Seminar, Univ. of Chicago                                        | May 2018                       |
| Algebra and Combinatorics Seminar, Loyola Univ                                      | Apr. 2018                      |
| CMI Seminar, Chennai Mathematical Institute                                         | Dec. 2017 & Jun. 2020 (online) |
| Algebraic Geometry Seminar, Washington University, St. Louis                        | Oct. 2017                      |

## ORGANISATION AND PROFESSIONAL SERVICES

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Study Advisor for masters students, Algebra, geometry and number theory, Leiden, Oct. 2025 - present

Member of the EWM 2024 - present

Co-organised workshop: *K3 surfaces & friends: Brauer groups and moduli*, Lorentz Center, Leiden, July, 2025  
<https://www.lorentzcenter.nl/k3-surfaces-en-friends-brauer-groups-and-moduli.html>

Masters’ study advisor for algebra group, Mathematisch Insstituut, Leiden starting from Nov 2025 -

Consultation (Overleg) team, Mathematisch Insstituut, Leiden Mar 2024 - present  
<https://www.universiteitleiden.nl/wiskunde-en-natuurwetenschappen/wiskunde/organisatie>

Co-organised *Intercity Geometry Seminar*, Amsterdam, Leiden, Utrecht, Antwerp, Summer semester 2024  
Informal intercity learning seminar: <https://www.math.leidenuniv.nl/~duttay/icspring24.html>

Co-organised: *Hodge Theory in Leiden*, a conference in honour of Chris Peters’ 75th birthday, Sep 5-6, 2024  
<https://www.math.leidenuniv.nl/~duttay/peters75.html>

Organised mini-symposium *Around Hyperkählers and K3s*, Leiden, December 6, 2023  
<https://www.math.leidenuniv.nl/~duttay/hksymposium23.html>

Co-organised the *Graduate Research Opportunities for Women (GROW@Bonn)*, 2022, 2023, 2024  
Conference aimed at female bachelor students for promoting math academia as a career.  
<https://www.mpim-bonn.mpg.de/GROW2024>

Served as tutor for *Algebraic Geometry Summer School - Hyperkähler and Prym varieties*, ICTP, July 2021.  
<https://indico.ictp.it/event/9610/other-view?view=ictp timetable>

Served as a recruitment committee member for *W3 professorship in mathematics*, Bonn, September 2020.

Reviewer *MathSciNet* (until July 2024), *zbMATH*.

Referee and quick opinion *Inventiones Mathematicae*, *Forum of Math (Sigma)*, *PRIMS Kyoto*.

Co-organised *MAGGC (Midwest Algebraic Geometry Graduate Conference)*, funded partially by the NSF,  
May 25-27 2018, University of Illinois at Chicago.

Co-organised *Algebraic Geometry Student Seminar*, Fall 2017 - Spring 2018, Northwestern University.

Served as a panel member for *TA seminar for 2nd year graduate students* Nov. 7, 2017, Northwestern University.

Served as a panel member for *graduate school applications* for NUAWM (Northwestern University Association for Women in Mathematics), an undergraduate club. Winter 2017, Northwestern University.

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## MISC. INFORMATION

Computer languages: Haskell, C, HTML, Latex.

Languages: Fluent in **English**, **Bengali** (native), **Hindi**.

General professional proficiency (Goethe-Zertifikat B2) in **German**.

Learning **Dutch**.